

a brief story of **ANOVA**

is it a test?

is it a model?

for **logistic regression**?!

...then **WHAT** is it?

ANOVA

is it a test?

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...then what is it?

→

→

→

No, it **USES** test statistic: F , χ^2

No, it **USES** a model under the hood

A special case of a general test for **MAIN EFFECTS**

1

Regression model → conditional function of the data

- *linear regression* - conditional mean,
- *logistic regression* - conditional probability (mean of Bernoulli),
- *quantile regression* - conditional quantile
- *Cox regression* - conditional hazard

Conditional function of
the response

$$\text{transformation}(fn(Y | X = x)) = X\beta$$

$$E(Y | X = x) = X\beta$$

$$\text{logit}(E(Y | X = x)) = X\beta$$

$$Q_i(Y | X = x) = X\beta$$

$$\lambda(Y | X = x) = X\beta$$

X – categorical predictors

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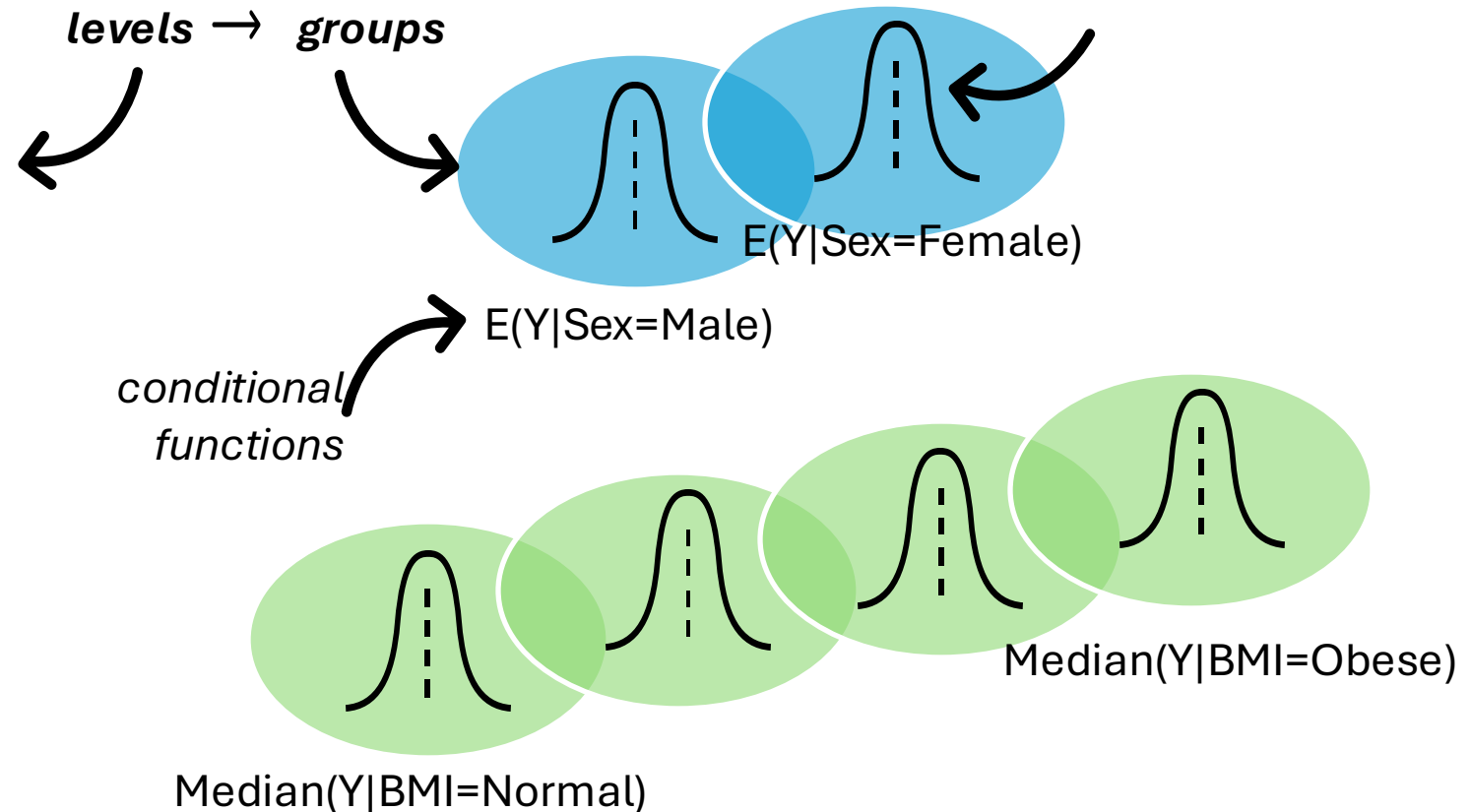
2

We may want to compare values of these conditional functions between groups

e.g. compare arithmetic means, medians, hazards, proportions (%)

factor Sex	Males
	Females

factor BMI	Underweight
	Normal
	Overweight
	Obesity



ANOVA

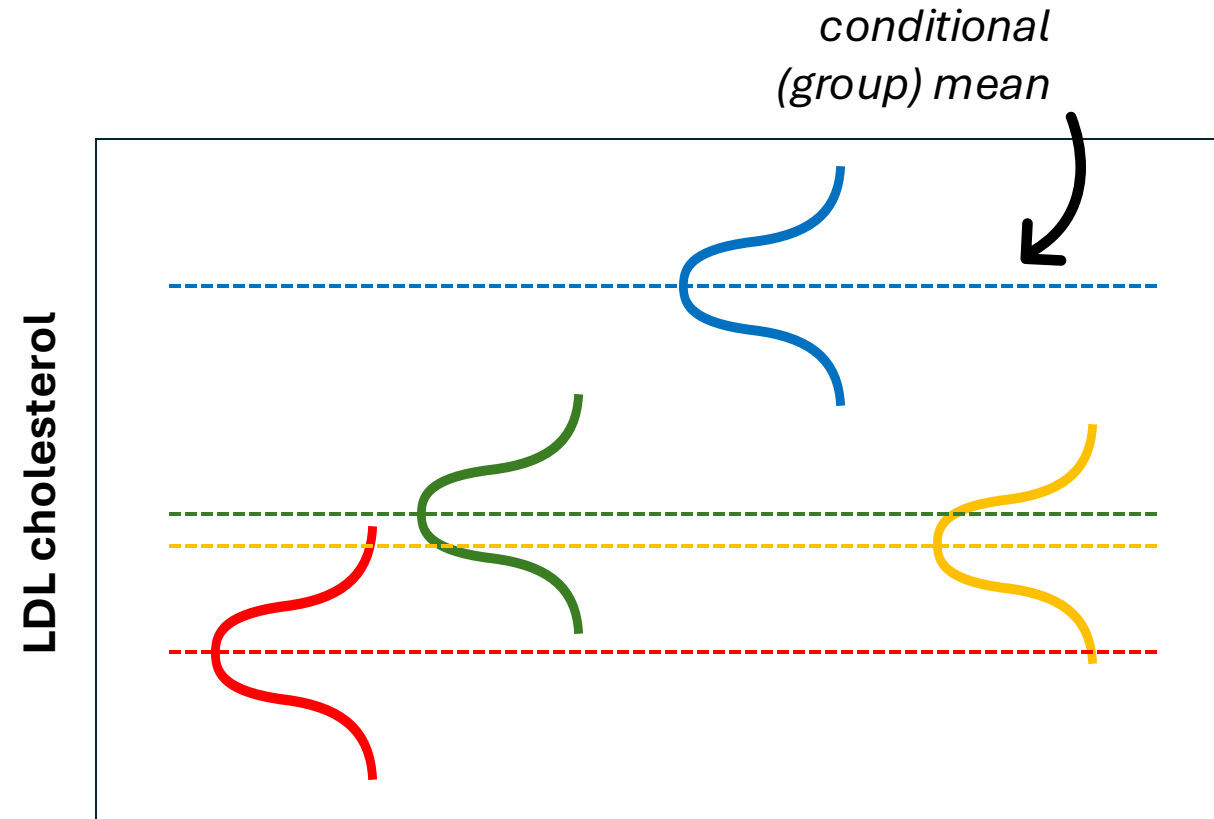
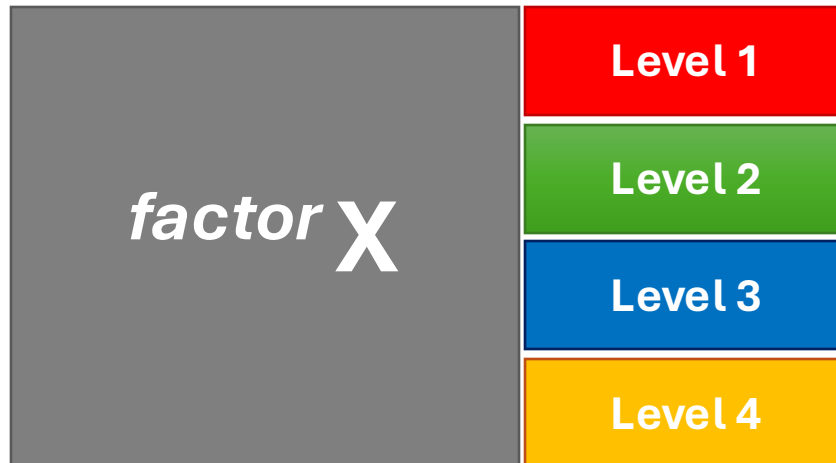
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3

Levels of categorical predictors (called **FACTORS**) form groups.

„Conditional” means, that the function is calculated within each group



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4

To compare them, we can fit appropriate model and test **CONTRASTS** of appropriate coefficients

Contrast = function (a combination) of model coefficients.

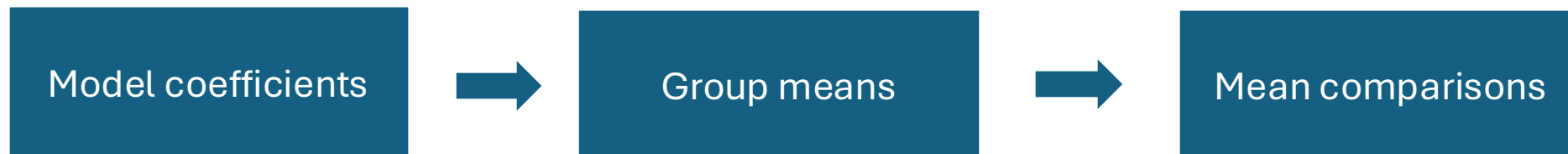
Interpretation of the coefficients depends on a model...

- **linear regression:** change in means
- **logistic regression:** change in log-odds or odds ratios (after exponentiation)
- **quantile regression:** change in selected quantile, e.g. medians

... and the contrast coding:

- **Treatment** – changes with respect to the selected reference (e.g. intercept)
- **Sum / Effect** – changes with respect to the global value, e.g. global mean

By appropriate constructing contrasts, we effectively compare –for example- group means:



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To compare them, we can fit appropriate model and test **CONTRASTS** of appropriate coefficients

For example (*treatment coding, general linear model = linear regression*):

(Intercept)	1.5	mean in the reference treatment, say A
Treatment_B	0.5	difference from the intercept, from A
Treatment_C	4.1	difference from the intercept, from A

Mean of A:	intercept = 1.5
Mean of B:	intercept + treatment_B = $1.5 + 0.5 = 2$
Mean of C:	intercept + treatment_C = $1.5 + 4.1 = 5.6$

	mean A	mean B	mean C	
Contrast for A:	1	0	0	
Contrast for B	1	1	0	
Contrast for C	1	0	1	
Contrast for B-A (subtract both above)	0	1	0	(makes sense! Treatment B is the difference from A!)
Contrast for C-B	0	1	1	(makes sense! C-A minus B-A = $C - A - B + A = C - B$)

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Such simple comparisons are called „**SIMPLE EFFECTS**”

legend: μ_{B1} – mean in the B1 group

Examples of:

- **1-way comparisons** (single factor A):
 μ_{A1} vs. μ_{A2} ,
- **2-way comparisons** (two factors A, B):
A=A1: μ_{B1} vs. μ_{B2} , **B=B2:** μ_{A1} vs. μ_{A2}
- **3-way comparisons** (two factors A, B, C):
A=A1, B=B2: μ_{C1} vs. μ_{C3} , **A=A2, C=C3:** μ_{B1} vs. μ_{B3}
- ... and any combination, e.g. μ_{A2B1} vs. **mean** ($\mu_{A1C2}, \mu_{A1B1} \dots$) – if provided they are meaningful...

This way we can test:

- 1) Planned contrasts (reflecting our research question)
- 2) Tukey's „all-pairwise” contrasts (whether it makes sense or not... often – it does not)
- 3) Dunnett's „all-vs.-baseline)

...answering questions that translate to comparisons between individual groups.

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But we may want to ask „globally”:

*do levels of the factor X
jointly ^(together) affect the value of the function
of the response variable?*

or simpler, for the classic ANOVA:

*do levels of the factor X
jointly affect the mean
of the response variable?*

Note, we don't ask for INDIVIDUAL differences, we just want to know, if the entire factor matters?

This is about MAIN EFFECTS.

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We need to test a very special set of contrasts now!

For a single factor A with 3 levels it will be:

$$\mu_{A1} - \mu_{A2}$$

and

$$(\mu_{A1} + \mu_{A2})/2 - \mu_{A3}$$

...and we need to test them „as a whole”, using a „joint test” rather than 2 separate tests! Look (example in R)!

```
> L
      [,1] [,2] [,3]
[1,]    0 -1.0    0    $\mu_{A1} - \mu_{A2}$ 
[2,]    0  0.5   -1    $(\mu_{A1} + \mu_{A2})/2 - \mu_{A3}$ 

> summary(glht(lm( response ~ group, data = data), L), test=Ftest())
```

General Linear Hypotheses

Global Test:

	F	DF1	DF2	Pr(>F)
1	174.6	2	87	3.471e-31

```
> summary(aov(lm( response ~ group, data = data)))
              Df Sum Sq Mean Sq F value Pr(>F)
group           2  293.75   146.88    174.6 <2e-16 ***
Residuals      87    73.17     0.84
```

```
test(contrast(emmeans(model, specs = ~group),
list(C1 = c(1, -1, 0), C2=c(.5, .5, -1))), joint = T)
df1 df2 F.ratio p.value
  2   87  174.626 <.0001
```

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This idea is not specific to just linear regression!

It applies TO ANY KIND OF REGRESSION MODEL!

If you have a set of estimated coefficients + the Variance-Covariance matrix
**then you can test any contrasts (simple effect)
and the main / interaction effects**

You can ask, for example if treatment affects

- the % of clinical successes
- the yearly rate of adverse reactions
- the 1st and 3rd quartile of the LDL-cholesterol
- the mean change in hemoglobin concentration
- the probability of obtaining lower scores in a Likert poll

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The contrasts for the main effects can be tested through the **Wald's inference**

But for **Likelihood-based models** we can employ also **Rao score test** and **Likelihood Ratio Test**

The Likelihood Ratio Test is used for comparison of nested models:

model without the main effect of interest

vs.

model with the main effect of interest

It uses the χ^2 statistic to check if **deviance** is reduced **ENOUGH** after adding the main effect of interest.

This is repeated for each model term (main effect), one by one.

Finally, it produces the ANOVA-like analysis called „ADOVA” = ***Analysis of Deviance***.

This is how we assess the impact of factors in various cases of the Generalized Linear Model (GLM):
the Poisson, logistic, multinomial, ordinal, negative-binomial, gamma, beta and other regression models.

We also remember, that the linear regression is a special kind of the GLM!

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We also remember, that the linear regression is a special kind of the GLM!

In this case, the „**deviance**” reduces to „**residual variance**”.

Here, the procedure uses **F statistic** to check, if the residual variance drops enough if you add the main effect of interest through the procedure of model comparisons.

**Now you know the meaning of „variance” in
„analysis of variance”! 😊**

```
> summary(aov(lm( response ~ group, data =  
data)))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	293.75	146.88	174.6	<2e-16 ***
Residuals	87	73.17	0.84		

```
> anova(lm( response ~ group, data = data),  
lm( response ~ 1, data = data), test="LRT")
```

Analysis of Variance Table

Model 1: response ~ group
Model 2: response ~ 1

	Res.Df	RSS	Df	Sum of Sq	Pr(>Chi)
1	87	73.17			
2	89	366.92	-2	-293.75	< 2.2e-16 ***

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<i>Type of model</i>	<i>Inference</i>	<i>Name</i>	<i>Interpretation</i>
Any model	Wald	Analysis of main effects + interactions	Depends on model
Likelihood-based (Generalized Linear Model)	Wald $\approx$ Rao $\approx$ LRT asymptotic.	Analysis of Deviance	Depends on model
General Linear Model	Wald = Rao = LRT = F	Analysis of Variance	Means

AN[C]OVA = analysis of main (+ interaction) effects

= do levels of categorical factors together impact the response enough?

Any kind of model

Cox, AFT, quantile, robust linear, GAM, and any other kind of a [fixed and mixed-effect] regression models



Generalized Linear Model

Logistic, multinomial, ordinal, Poisson, negative binomial, gamma [fixed & mixed-effects] regressions
Fitted using Maximum Likelihood or GEE



General Linear Model

Fitted using OLS, GLS (for unequal variances and dependency) or GEE (also non-normal data) + linear (Gaussian) mixed models

Assesses jointly contrasts (functions) of coefficients of an underlying model with categorical (+optionally numerical) predictor variables. *Uses Wald's approach for testing the contrasts.*

In R: car::Anova(), anova(), emmeans::joint_tests()

Analysis of Deviance

The „impact” of the factors is assessed through the reduction of deviance for each factor added to the model. Both Wald's and the LRT approach are used to compare group means (transformed via link-function) but may vary in outcomes.

In R: anova(), car::Anova(), emmeans::joint_tests()

Analysis of Variance

Reduction of deviance simplifies to reduction of residual variance – thus the name. Group means are compared through the Wald's and (equivalently) the LRT approaches.

In R: anova(), car::Anova(), emmeans::joint_tests(), aov()

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Is ANOVA a test?

No. **It uses a series of Chi2 or F tests** to assess the impact of each factor (and their interactions)

Is ANOVA a model?

No. **It operates over a model under the hood**, assessing its main and interaction effects.

All assumptions of ANOVA come from the underlying OLS estimation procedure!

- Normality of residuals → normality per group
- Heteroscedasticity of residuals → per-group homogeneity of variance
- Independence of residuals → independent observations

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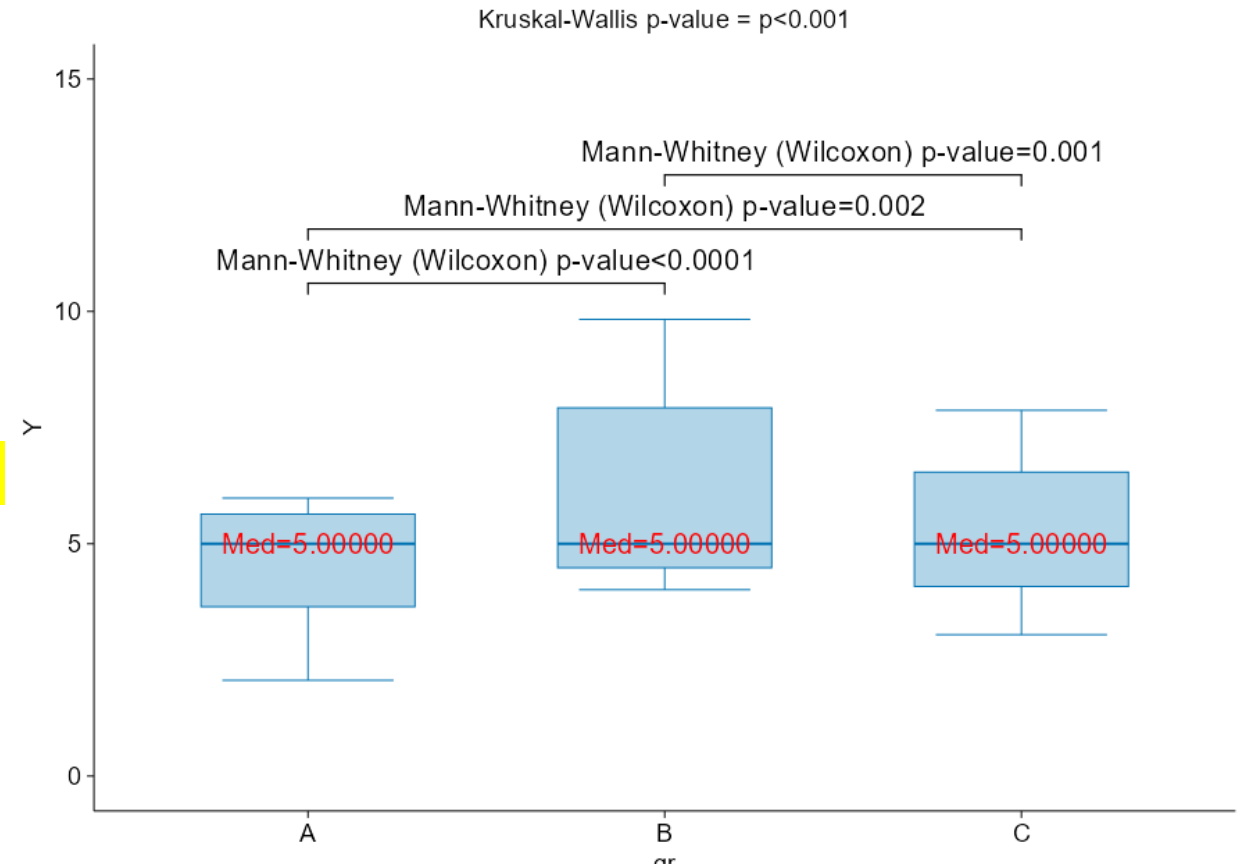
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What to do if these assumption fail?

Use a more generalized parametric model!

DO NOT switch automatically to non-parametric rank-based methods.

They will entirely change your null hypotheses to something called „stochastic superiority”, which can mislead you! →



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What to do if these assumption fail?

Use a more generalized (semi) parametric model preserving the H_0 interpretation in terms of arithmetic means!

Assumption violated	Use instead	Comment
Homogeneity of variance	Generalized Least Square (GLS)-fit linear model with same terms as originally. You can ALWAYS start with it, no need for testing Levene/Bartlett!	Have to specify the covariance structure. You can start with <i>unstructured</i> and hope the model fits. Otherwise try other structures.
Independence (e.g. repeted observations, clustered data)	- Generalized Least Square, - Generalized Estimating Equations (GEE) - Linear Mixed-Effect Model	GLS - as above GEE - only asymptotic inference LMM - play with random effects
Non-normality	- Generalized Estimating Equations (GEE) - Permutation ANOVA	GEE - only asymptotic inference Exhaustive computation
Outliers	Robust linear (mixed) model	Exhaustive computation

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Example: **heterogeneity of variance** – see? **NO NEED FOR KRUSKAL-WALLIS!**

```
> set.seed(100)
> d <- data.frame(group = factor(rep(LETTERS[1:3], each=30)),
+               response = c(rnorm(30, mean=1, sd=1),
+               rnorm(30, mean=2, sd=2),
+               rnorm(30, mean=3, sd=5)))
>
> fitols <- lm(response ~ group, data = d)
> fitgls <- gls(response ~ group, weight = varIdent(form = ~1 | group), data = d)

> summary(aov(fitols))
              Df Sum Sq Mean Sq F value Pr(>F)
group           2   47.6   23.80   1.995  0.142
Residuals      87 1038.0   11.93

> joint_tests(fitgls, mode="df.error")
  model term df1  df2 F.ratio p.value
group           2   85   4.733  0.0113 # smaller DF → covariance estimates
```



functions to analyse main effects

General Linear Model:

ANOVA, ANCOVA

Generalized Linear Model:

Analysis Of Deviance

Other models:

Analysis Of Main Effects

Function	Type of analysis	Models	Way of testing	Other capabilities
aov()	Type-1 ⚠	General Linear Model (linear regressio)	Wald	Supports repeated observations and more complex designs via Error() term
anova()	Type 1 / 3	Virtually any kind of model, also GLM, Cox, mixed that expose certain interface. Typically used to compare models, NOT typical ANOVA.	Wald, LRT, Rao — depends on compared models	Supports everything that the author of certain model implementation provided. For example: for some models it can do marginal (type-3) or sequential (type-1), can use LRT or Rao testing.
car::Anova()	Type 2 / 3	Virtually any kind of model, also GLM, Cox, mixed that expose some interface		As above. For type-3 analysis requires contrasts set to „contr.sum” ⚠
emmeans:: joint_tests()	Type 3 (rarely 2)	Any model that emmeans supports, including GLS, GEE, mixed models, quantile regression, robust models, survival models, ordinal regressions, entire GLM, and virtually ANY via „qdrq()” interface https://tinyurl.com/emmeans-models	Wald	Supports everything the used model does (e.g. correlated data, skewness, heteroscedasticity, censoring, etc). Doesn't require setting contrasts! Can use advanced degrees of freedom like Satterthwaite, Kenward-Roger, etc. Can use robust estimators of variance. Deals with EM-means, NOT THE RAW MEANS ⚠ (weighting can be changed)



One of the biggest WARNINGS in R

Type-1 sum-of-square ANOVA **aov()** made default

How many research must have been flawed because of the fact that...

...the results *DEPEND ON THE ORDER* of variables entered

```
> summary(aov(PD ~ Method + PD_Bas, data=data))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Method	1	3.75	3.750	3.429	0.0692 .
PD_Bas	1	21.77	21.766	19.903	3.89e-05 ***
Residuals	57	62.33	1.094		

```
> summary(aov(PD ~ PD_Bas + Method, data=data))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
PD_Bas	1	10.65	10.654	9.742	0.002826 **
Method	1	14.86	14.862	13.590	0.000509 ***
Residuals	57	62.33	1.094		

This is a simple ANCOVA. Method is balanced (N=30 per method). No missing data.
But the PD_Bas is correlated a bit with the PD ($R=0.35$) **and this is enough to cause problems.**

Type-2 ANOVA has no such problem attached:

```
> car::Anova(lm(PD ~ Method + PD_Bas, data=data), type="2")
```

Anova Table (Type II tests)

Response: PD

	Sum Sq	Df	F value	Pr(>F)
Method	14.862	1	13.590	0.0005085 ***
PD_Bas	21.766	1	19.903	3.887e-05 ***
Residuals	62.334	57		

```
> car::Anova(lm(PD ~ PD_Bas + Method, data=data), type="2")
```

Anova Table (Type II tests)

Response: PD

	Sum Sq	Df	F value	Pr(>F)
PD_Bas	21.766	1	19.903	3.887e-05 ***
Method	14.862	1	13.590	0.0005085 ***
Residuals	62.334	57		